Tank 9: Improve one feature

Goal: I can use test evidence to change one feature and check whether it helps.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 16, grade 6 path; Playbook B meeting 17, grade 6 path

Ask | Plan | Build | Test | Improve

You are done when

Show one change you made after testing. Compare the results before and after.

You will use

- Clean cardboard
- Paper
- Pencils
- Rulers
- Student scissors
- Masking tape
- Reusable trays
- Prototype
- Previous test record
- Approved test objects

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Read today's goal.
- Look at the example.

2 Step 2: Tell your partner — 1 min

- Tell your partner one useful detail.

3 Step 3: Watch and try — 5 min

- Compare a design change with a decorative change.
- Trace one failed test to the part causing it.

4 Step 4: Make it work — 17 min

- Choose one problem from your test record.
- Change the relevant shape, connection, or movement.
- Repeat the same three-trial test.

5 Step 5: Check one thing — 5 min

- Compare the old and new results.
- Keep the improvement or explain why you will undo it.
- Record what the evidence suggests next.

6 Step 6: Reset together — 3 min

- Return tools and reusable pieces.
- Label your work with the class storage code.
- Clear your table and floor.

7 Step 7: Show your result — 2 min

- Show one change you made after testing. Compare the results before and after.

What to turn in

Before and after results for one change.

No computer? Another way to do it

Draw the change and explain the result you expect.

Finished early? Next challenge

Improve the same work or rehearse your explanation. Ask before taking more materials.

Stay safe

Use teacher-approved materials and tools. Keep paths clear. Return tools before moving.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.